VERY SHORT RANGE AIR DEFENCE SYSTEM

5,175 missiles; **$1.5 billion** (Rs 11,213.1 crore)

155mm TOWED GUN SYSTEM

400 guns; **Rs 4,000 crore**

SELF-PROPELLED AIR DEFENCE GUN SYSTEM

104; **$2.4 billion** (Rs 17,941 crore)

SIG 716 RIFLES

72,400 rifles; Upwards of **Rs 700 crore**

QUICK REACTION SURFACE-TO-AIR MISSILES

56 mobile launcher systems; **$1.5 billion**



AIRBUS

Hanwha Defence but have progressively been indigenised by L&T in a specially made assembly line in Hazira, Gujarat. The K9 order is likely to be announced during the MoD's biennial DefExpo to be held between March 10 and 13 this year.

The MoD directive had its origin in meetings with the PM in November 2021. The presentation made by then secretary (defence production) Raj Kumar outlined why the Aatmanirbhar Bharat policy was a slow starter. Kumar had successfully carried out the MoD's largest restructuring—cracking up the four-decade old Ordnance Factory Board and 41 factories to turn it into seven state-owned corporations. The presentation outlined how despite the MoD's 'positive indigenisation list'—a ban on importing 351 items since 2020—the services had not placed a single order on an Indian manufacturer. Government-owned defence public sector undertakings (DPSUs) were spending billions of dollars importing components for indigenous defence systems.

On January 6, the MoD Defence Procurement Board (DPB) began drawing up a 'kill list' of imported equipment, hacking away at deals for helicopters, missiles and assault rifles which would not add to India's technology base. On January 14, the Indian army placed a $20 million (Rs 149.5 crore) order with private sector drone-maker Idea Forge, the largest such order on a private firm.

Lt Gen Subrata Saha, former deputy chief of the army staff and member of the National Security Advisory Board (NSAB), predicts a threefold jump in private sector defence orders over the next few years. "Until July last year, the total defence business was Rs 80,000 crore, with Rs 63,000 crore coming from the DPSUs and Rs 17,000 crore from the private sector. The government has set a target for Rs 1.25 lakh crore by 2025. The increase of Rs 45,000 crore will mainly come from the private sector," he says.

THE MoD CALLS FOR SUBSTANTIAL REFORMS IN THE DRDO AND AN EXAMINATION OF FOREIGN COMPONENTS IN DRDO EQUIPMENT

Private sector firms say they need R&D funding to field indigenous products. "We don't have the business volumes to be able to invest more.... If the government gave the private sector a fraction of the Rs 21,000 crore it spends on the DRDO, it will transform Indian industry," says the CEO of a top defence firm.

Beijing's transformation has been backed by years of a focused government approach to capture defence technology, aided, no doubt, by what a 2020 Belfer Center paper calls "Chinese intellectual property theft in the range of $180 to $540 billion per year". Chinese state entities are believed to have used cyber espionage to vacuum detailed production designs from the West for use by their indigenous industry. India's approach to acquiring defence technology, as a private sector defence industry CEO puts it, has been more "Gandhian". It is unlikely that the Indian government will sign off on a grand IPR (intellectual property right) theft. What India needs instead, says the official, is comprehensive ease of doing business for start-ups and tax breaks. "The MoD needs to understand that there is a lack of experience, and our domain knowledge in several key areas is still at nascent stages. It is essential to hire overseas experts. The Japanese, Germans, Koreans and Chinese all followed this route, but here the hurdles make it impossible," says the CEO of a private sector start-up.

The few attempts at breaking free have yielded admirable results in niche technology areas. Four years ago, the IAF dangled a carrot of a Rs 100 crore order for a Meher Baba swarm drone contest. The IAF spent Rs 25 crore to handhold nearly 57 firms that began developing prototypes. The contest seeded an entire swarm drone ecosystem. It created multiple start-ups, like NewSpace Research and Tech which bagged a Rs 200 crore swarm drone order from the army last year, and saw the emergence of hungry young start-ups like Veda Defense Systems Pvt Ltd and Dhaksha Unmanned Systems. But these industries need orders if they are to grow.

L&T's K9 howitzers, for instance, created a supply chain of about 1,100 industrial partners, mostly MSMEs (micro, small and medium enterprises). These have delivered over 80 per cent indigenous work packages by locally producing over 14,000 types of components per howitzer and have achieved over 50 per cent indigenisation (by value).

In his 1990 book, *The Militarisation of Mother India*, defence analyst Ravi Rikhye noted how Chinese weaponry was then terribly crude, but entirely Chinese. 'No one can twist the Chinese tail. Military power, like any other power, cannot be borrowed. It has to be acquired from one's strength. Otherwise, we become another's surrogate,' he wrote. A compelling reason as any for why the government's newest initiative must succeed. ■